VIT BHOPAL
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Assignment -1

Introduction to Calculus **Subject Code: MAT1031**
Fall Semester -2025-2026 **Slot: A11+A12+A13**

Please ensure that your name, registration number, subject, and subject code are clearly mentioned on the front page. Submit the soft copy in the proper format by 06.10.25.

1. In a space curve Let $w = x^2e^{2y}$. Find the value of dw/dt at the point $(l, \ln 2)$ on the curve $x = cost,\ y = log(t + 2),$.
2. Given $x^2 + \cos y + z^3 = 1$, find $\frac{\partial z}{\partial y}$.
3. If $u = r^m, r = \sqrt{x^2 + y^2 + z^2}$ find the value of $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}$
4. For the function $f(x, y) = xy^2 + e^{x^2y}$ show $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$.
5. Show that $u(x, y) = log(x^2 + y^2)$ satisfies the partial differential equation:
$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$
6. If $y = sin(sin\,x)$, prove that $y_2 + y_1 \tan x + y\, cos^2 x = 0$.
7. Find y_n, if $y = x^2 \log x$.
8. Find the nth differential coefficient i.e., y_n of $y = \ln(ax + b)$.
9. If $A = \pi x^2 - 20\,x$. Calculate the rate of change of A with respect to x.
10. Differentiate $y = \frac{x^3}{1+x^3}$ with respect to x.
11. If $u = tan^{-1}\left(\frac{x}{y}\right)$, find $\frac{du}{dx}$, where $x^2 + y^2 = a^2$.
12. Use Taylor's formula for $f(x, y)$ at the origin to find quadratic approximations of f near the origin.
    i) $y \sin x$ ii) $e^x \ln(1 + y)$ iii) $\ln(2x + y + 1)$
13. Expand $f(x, y) = e^{xy}$ in Taylors Series at (1, 1) upto second degree.
14. Examine the maxima and minima of the following functions.
    $(i)\ x^3 - y^3 - 3xy$
    $(ii)\ x^3 + 3xy^2 - 15x - 12y$
    $(iii)\ f(x, y) = x^3y^2(a - x - y)$
15. Find the maxima and minima of $x^4 + y^4 - 2x^2 + 4xy - 2y^2$.

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16. flat circular plate is heated so that the temperature at any point (x, y) is $u(x, y) = x^2 + 2y^2 - x$. Find the coldest point on the plate.
17. The temperature $u(x, y, z)$ at any point in space is $u = 400xyz^2$ .Find the highest temperature on surface of the sphere $x^2 + y^2 + z^2 = 1$.
18. Find the minimum values of x^2yz^3 subject to the condition $2x + y + 3z = a$.
19. Find the shortest and the longest distances from the point $(1,2,-1)$ to the sphere $x^2 + y^2 + z^2 = 24$, using Lagrange's method of constrained maxima and minima.
20. Find the maximum and minimum value of $x^2 + y^2 + z^2$ subject to the condition $x + y + z = 3a$.
21. Find the following integrals:

$(i)\ \int sin^3x \cos^2x\ dx$ $\qquad$ $(ii)\ \int \frac{1}{1+\tan x} dx$

$iii)\ \int sin^3x\ \ dx$ $\qquad$ $iv)\ \int \sin 3x \cos 4x\ \ dx$

$v)\ \int \tan^4 x\ \ dx$ $\qquad$ $vi)\ \int \sin 4x \sin 8x\ \ dx$